Apache Nutch Search Engine Assignment

Submitted By

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Abstract
This report explores the configuration and deployment of Apache Nutch, an open-source web crawler and search engine. Nutch provides a scalable solution for crawling web content and indexing it for search purposes. The project aimed to set up Nutch for crawling a specific domain, configuring it to efficiently gather data from web pages. The report outlines the technical setup, including the use of Apache Tomcat for deployment, and discusses how Nutch can be used to create functional web search applications.

Introduction
Apache Nutch is a robust and scalable web crawler and search engine platform designed to index and search web content across a wide range of domains. As part of this project, the focus was on configuring Nutch to gather data from a designated domain and deploy it using Apache Tomcat. The goal was to implement a functional web crawling system that collects data from web pages and makes it searchable. This report delves into the process of setting up Nutch, configuring the required environment, and deploying the application, while also highlighting the relevance of Nutch for building search applications in various domains.

Project Overview

Software Used:

1. Apache Nutch - Version 0.9
2. Apache Tomcat - Version 8.5.95
3. Java Development Kit (JDK) - Version 21.0.1
4. Cygwin 64 Environment

Objective:
The objective was to crawl a specific domain (http://www.nits.ac.in) and collect web content for indexing. The output was a fully functioning crawler that stored retrieved data and a search engine interface that allowed users to query the indexed content.

Implementation

Environment Setup:
1. Set up Java by configuring the environment variables, setting paths, and JAVA_HOME.
2. Extract Nutch archive and configure properties:
   - Make bin/nutch executable using chmod.
   - Create a seed list of URLs to crawl (in this project, list.txt is created).
   - Update nutch-site.xml and configure regular expressions for URL filters.

3. Start crawling using the command in the “bin”: ./nutch crawl urls -dir Crawl. (This step creates a “Crawl” file that contains the retrieved data of the mentioned URL.

Deploying Nutch in Tomcat:

1. Deploy the Nutch WAR file in Apache Tomcat.
2. Access Tomcat Manager and upload the nutch-0.9.war file.
3. Got to C:\ProgramFiles\Apache Software Foundation\Tomcat 8.5\webapps\nutch-0.9\WEB-INF\classes and edit searcher.dir in nutch-site.xml to point to the crawl directory.
4. In the nutch-site.xml, perform the following modifications:

```
<configuration>

<property>

  <name>http.agent.name</name>

  <value>SaeeNutch</value>

</property>

</configuration>
```

5. Edit the search.jsp file to cope up with the escape sequence error.
6. Restart Tomcat and test search functionality by visiting http://localhost:8080/nutch-0.9.

Nutch Architecture and Workflow

Apache Nutch’s architecture is modular and distributed, involving several components:

1. Crawler - Downloads web pages from the specified URLs.
2. Parser - Extracts relevant content from the downloaded pages.
3. Indexer - Stores the extracted content in an index for future queries.
4. Query Processor - Handles search queries from users and returns relevant indexed results.

This modular architecture enabled us to crawl, parse, and search the web content efficiently.

Conclusion

This project demonstrates the successful deployment of Apache Nutch as a domain-specific web crawler and search engine. Despite its complexity and challenges in configuration, performance and documentation, Nutch proved to be a cost-effective and scalable solution for indexing web content. With proper configuration, it can be tailored to various applications such as general-purpose search engines, enterprise search, or vertical search engines. In future work, we aim to optimize Nutch's performance and explore its integration with big data platforms like Hadoop for more efficient processing of large datasets.

References

https://cwiki.apache.org/confluence/display/NUTCH/NutchTutorial

https://nutch.apache.org/documentation/tutorials/

https://tomcat.apache.org/tomcat-8.5-doc/changelog.html

Results

By the end of the project, we successfully created a search engine capable of crawling and indexing content from the designated domain. Users could input search queries, and the engine would return relevant results based on the indexed content.

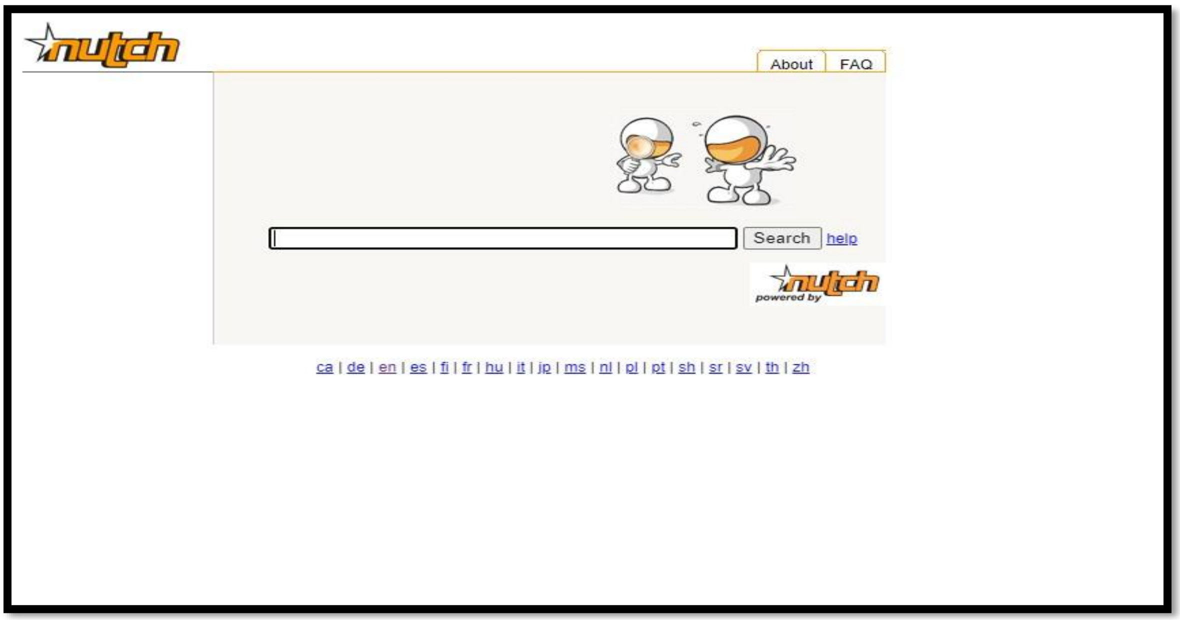

Fig 1: Nutch Manager App – Home Page

Fig 2: On searching “artificial” all the results on the URL are shown.